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B.Tech Artificial Intelligence & Data Science - 2nd Semester

Object Oriented Programming – Lab Assignment 12

Exercises with Experiments

Exercise 1: Component Positioning

Code:

```
import javax.swing.*;
public class Ex1_Positioning{
    public static void main(String[] args){
        JFrame frame = new JFrame("Excercise 1");
        frame.setLayout(null);
        JLabel l1 = new JLabel("Label 1");
        JLabel l2 = new JLabel("Label 2");
        JLabel l3 = new JLabel("Label 3");
        l1.setBounds(50,50,100,30);
        l2.setBounds(150,100,100,30);
        l3.setBounds(250,150,100,30);
        frame.add(l1);
        frame.add(l2);
        frame.add(l3);
        frame.setSize(400,300);
        frame.setVisible(true);
    }
}
```

Output:

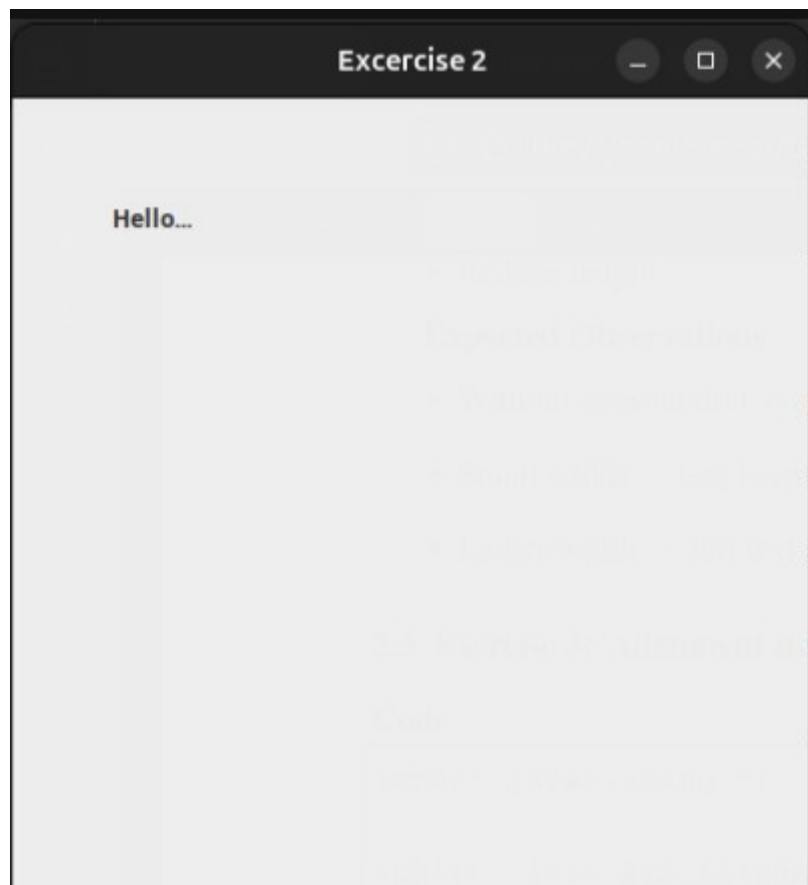


Exercise 2: Visibility and Size

Code:

```
import javax.swing.*;
public class Ex2_visibility{
    public static void main(String[] args){
        JFrame frame = new JFrame("Excercise 2");
        frame.setLayout(null);
        JLabel label = new JLabel("Hello World !");
        label.setBounds(50,50,40,20);
        frame.add(label);
        frame.setSize(400,400);
        frame.setVisible(true);
    }
}
```

Output:



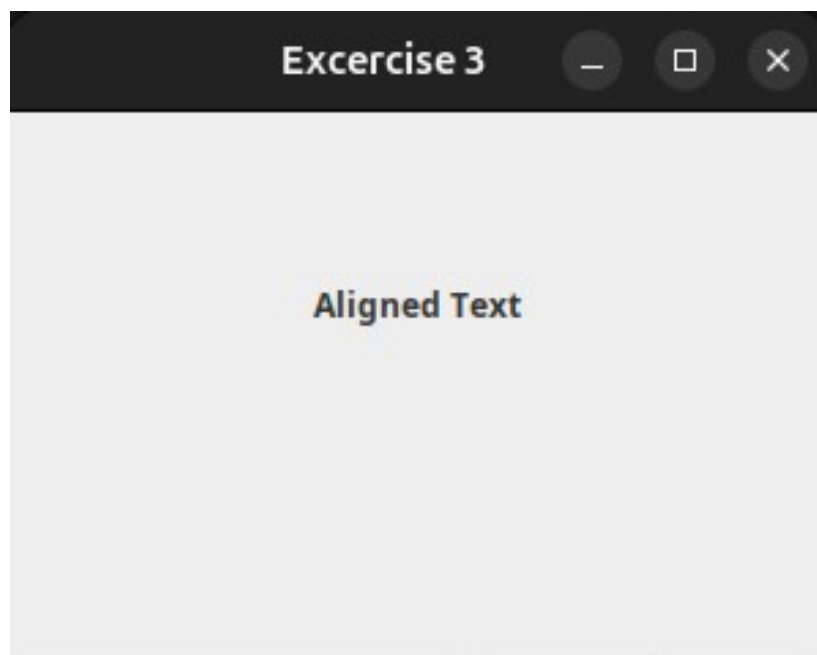
Exercise 3: Alignment Behavior

Code:

```
import javax.swing.*;
public class Ex3_Alignment{
    public static void main(String[] args) {
        JFrame frame = new JFrame("Excercise 3");
        frame.setLayout(null);

        JLabel label = new JLabel("Aligned Text");
        label.setBounds(50,50,200,40);
        label.setHorizontalAlignment(JLabel.CENTER);
        frame.add(label);
        frame.setSize(300,200);
        frame.setVisible(true);
    }
}
```

Output:



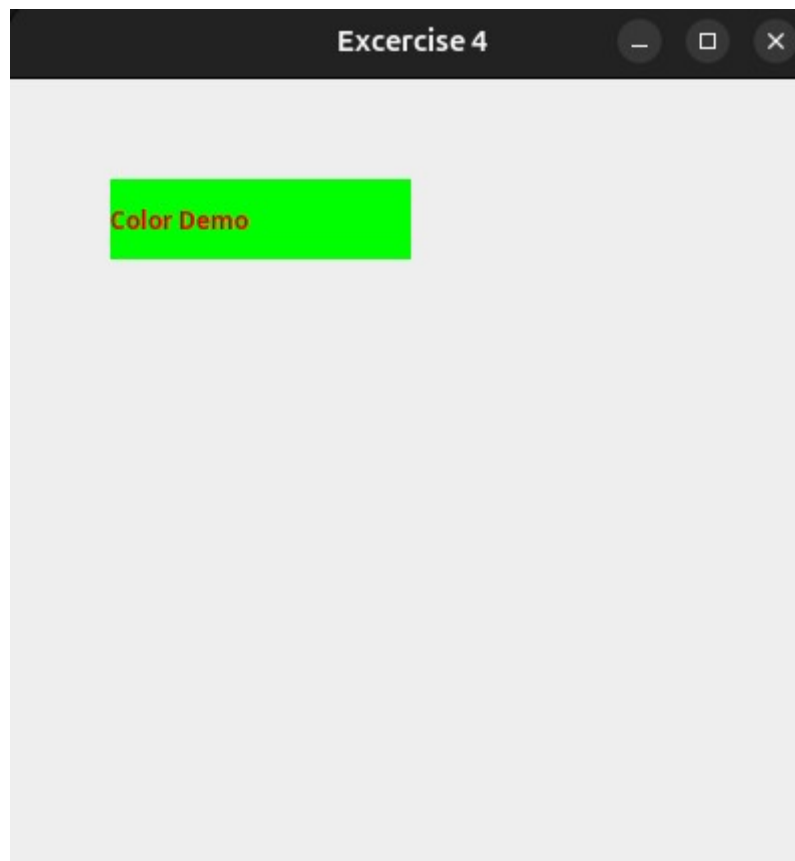
Exercise 4: Color and Background

Code:

```
import javax.swing.*;
import java.awt.*;

public class Ex4_color{
    public static void main(String[] args){
        JFrame frame = new JFrame("Excercise 4");
        frame.setLayout(null);
        JLabel label = new JLabel("Color Demo");
        label.setBounds(50,50,150,40);
        label.setForeground(Color.RED);
        label.setBackground(Color.GREEN);
        label.setOpaque(true);
        frame.add(label);
        frame.setSize(400,400);
        frame.setVisible(true);
    }
}
```

Output:



Application Integration – Scenarios

Scenario 1: Student List (Basic)

Code:

```
import java.util.*;
class Student{
    String name;
    Student(String name){
        this.name=name;
    }
    String getName(){
        return name;
    }
}
class StudentSystem{
    ArrayList<Student> students = new ArrayList<>();
    void addStudent(Student s){
        students.add(s);
    }
    void showStudents(){
        for(Student s : students){
            System.out.println(s.getName());
        }
    }
}
```

Scenario 2: Attendance System (Intermediate)

Code:

```
import java.util.*;
class Student2{
    String name;
    boolean present;
    Student2(String name){
        this.name=name;
        present=false;
    }
    String getName(){
        return name;
    }
    void markPresent(){
        present=true;
    }
    void markAbsent(){
        present=false;
    }
    boolean isPresent(){
```

```

        return present;
    }
}
class AttendanceSystem{
    ArrayList<Student2> students = new ArrayList<>();
    void addStudent(Student2 s){
        students.add(s);
    }
    void showStudents(){
        for(Student2 t : students){
            System.out.println(t.getName());
        }
    }
    void markPresent(Student2 name){
        for(Student2 t : students){
            if(t.getName().equals(name)){
                t.markPresent();
            }
        }
    }
    void showPrsentStudents(){
        for(Student2 std : students){
            if(std.isPresent()){
                System.out.println(std.getName());
            }
        }
    }
}
}

```

Scenario 3: Quiz Participation Tracker

Code:

```

import java.util.*;

class Participant{
    String name;
    boolean attempted;
    Participant(String nm){
        this.name=nm;
        this.attempted=false;
    }
    String getName(){
        return name;
    }
    void markAttempted(){
        attempted=true;
    }
    void markNotAttempted(){
        attempted=false;
    }
    boolean didAttempt(){
        return attempted;
    }
}

```

```

    }
}
class QuizSystem{
    ArrayList<Participant> part = new ArrayList<>();
    void addParticipant(Participant p){
        part.add(p);
    }
    String markAttempted(String name){
        for(Participant p : part){
            if(p.didAttempt()){
                return("Has already attempted");
            }
            p.markAttempted();
            return name + " marked as attempted";
        }
        return "Participant not found";
    }
    String markNotAttempted(String name){
        for(Participant p : part) {
            if (p.getName().equals(name)) {
                if (!p.didAttempt()) {
                    return ("ALready not attempted");
                }
                p.markNotAttempted();
                return name + "marked not attempted";
            }
        }
        return "Participant not found";
    }
    String showAttempted(){
        String result = "";
        for(Participant p : part){
            if(p.didAttempt()){
                result+=p.getName()+"\n";
            }
        }
        if(result.equals("")){
            return "No participants attempted";
        }
        return result;
    }
}
}

```

```

import javax.swing.*.*;
public class UI{
    public UI(QuizSystem sys){
        JFrame frame = new JFrame("Quiz Tracker");
        frame.setLayout(null);
        JTextField input = new JTextField();
        input.setBounds(50,30,200,30);

        JTextArea output = new JTextArea();
        output.setBounds(50,80,300,120);
    }
}

```

```

JButton show = new JButton("Show Attempted");
JButton mark = new JButton("Mark Attempted");
JButton unmark = new JButton("Mark Not Attempted");

show.addActionListener(e->
    output.setText(sys.showAttempted()));
mark.addActionListener(e->
    output.setText(sys.markAttempted(input.getText())) );
unmark.addActionListener(e->
    output.setText(sys.markNotAttempted(input.getText())) );
frame.add(input);
frame.add(show);
frame.add(mark);
frame.add(unmark);
frame.add(output);
frame.setSize(420,300);
frame.setVisible(true);
frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

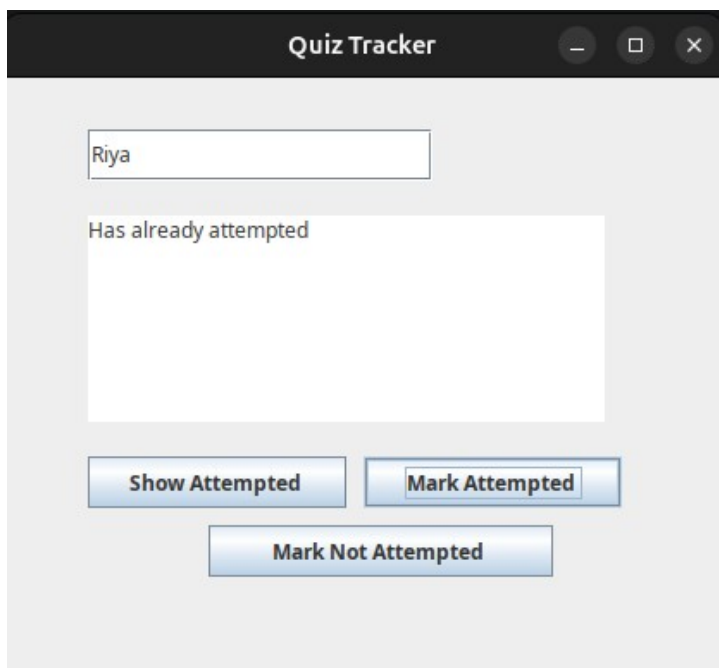
```

```

public class MainApp{
    public static void main(String[] args){
        QuizSystem sys = new QuizSystem();
        sys.addParticipant(new Participant("Amit"));
        sys.addParticipant(new Participant("Riya"));
        sys.addParticipant(new Participant("John"));
        new UI(sys);
    }
}

```

Output:



Application Scenario - STUDENT TASK

Code:

Superhero.java :

```
public class Superhero {  
    String name;  
    boolean onMission;  
  
    Superhero(String n) {  
        name = n;  
        onMission = false;  
    }  
  
    void sendOnMission() {  
        onMission = true;  
    }  
  
    void recallToBase() {  
        onMission = false;  
    }  
  
    boolean isOnMission() {  
        return onMission;  
    }  
  
    String getName() {  
        return name;  
    }  
}
```

MissionControl.java :

```
import java.util.*;  
  
public class MissionControl {  
    ArrayList<Superhero> heroes = new ArrayList<>();  
  
    void addHero(Superhero h) {  
        heroes.add(h);  
    }  
  
    String sendHero(String name) {  
        for (Superhero h : heroes) {  
            if (h.getName().equalsIgnoreCase(name)) {  
                if (h.isOnMission()) {  
                    return name + " is already on a mission!";  
                }  
                h.sendOnMission();  
                return name + " has been sent on a mission.";  
            }  
        }  
    }  
}
```

```

    }
}
return "Superhero " + name + " not found!";
}

String recallHero(String name) {
    for (Superhero h : heroes) {
        if (h.getName().equalsIgnoreCase(name)) {
            if (!h.isOnMission()) {
                return name + " is already at the base!";
            }
            h.recallToBase();
            return name + " has been recalled to base.";
        }
    }
    return "Superhero " + name + " not found!";
}

String showActiveMissions() {
    String result = "";
    for (Superhero h : heroes) {
        if (h.isOnMission()) {
            result += h.getName() + "\n";
        }
    }
    if (result.equals("")) {
        return "No superheroes are currently on a mission.";
    }
    return "Heroes on Active Missions:\n" + result;
}
}

```

UImain.java :

```

import javax.swing.*.*;

public class UImain {

    public UImain(MissionControl sys) {
        JFrame frame = new JFrame("Superhero Mission Control");
        frame.setLayout(null);

        // Label for input
        JLabel inputLabel = new JLabel("Enter Superhero Name:");
        inputLabel.setBounds(50, 20, 200, 25);

        // Input field
        JTextField input = new JTextField();
        input.setBounds(50, 50, 200, 30);

        // Output area
        JTextArea output = new JTextArea();
        output.setBounds(50, 100, 300, 120);
    }
}

```

```

output.setEditable(false);
output.setLineWrap(true);

// Buttons
JButton showBtn = new JButton("Show Active Missions");
JButton sendBtn = new JButton("Send on Mission");
JButton recallBtn = new JButton("Recall to Base");

showBtn.setBounds(50, 240, 180, 30);
sendBtn.setBounds(50, 280, 180, 30);
recallBtn.setBounds(50, 320, 180, 30);

// Button Actions
showBtn.addActionListener(e ->
    output.setText(sys.showActiveMissions())
);

sendBtn.addActionListener(e ->
    output.setText(sys.sendHero(input.getText().trim()))
);

recallBtn.addActionListener(e ->
    output.setText(sys.recallHero(input.getText().trim()))
);

// Add all components
frame.add(inputLabel);
frame.add(input);
frame.add(output);
frame.add(showBtn);
frame.add(sendBtn);
frame.add(recallBtn);

frame.setSize(420, 400);
frame.setVisible(true);
frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
}

```

MainAppFile.java :

```

import javax.swing.*.*;
public class MainAppFile {
    public static void main(String[] args) {
        MissionControl sys = new MissionControl();
        sys.addHero(new Superhero("Batman"));
        sys.addHero(new Superhero("Spiderman"));
        sys.addHero(new Superhero("WonderWoman"));

        // Build GUI and connect system
        SwingUtilities.invokeLater(() -> new UImain(sys));
    }
}

```

Output:

Superhero Mission Control

Enter Superhero Name:

spiderman

Heroes on Active Missions:

Batman
Spiderman

Show Active Missions

Send on Mission

Recall to Base

Superhero Mission Control

Enter Superhero Name:

spiderman

spiderman has been recalled to base.

Show Active Missions

Send on Mission

Recall to Base

Superhero Mission Control

Enter Superhero Name:

WonderWoman

WonderWoman has been sent on a mission.

Show Active Missions

Send on Mission

Recall to Base

Superhero Mission Control

Enter Superhero Name:

WonderWoman

Heroes on Active Missions:

Batman
WonderWoman

Show Active Missions

Send on Mission

Recall to Base